Q1

1. (a) use the model

$$Y_{it} - \bar{Y}_i = \beta_1 (X_{it} - \bar{X}_i) + (u_{it} - \bar{u}_i)$$

because α_i only different between villiges, taking mean on different villiges during 2016 to 2019 can eliminate α_i

(b) the soil's quality

2.

(a) use a mode

$$Y_{it} = \beta_1 X_{it} + \delta_{2016} D_{2016} + \delta_{2017} D_{2017} + \delta_{2018} D_{2018} + \beta_0 + u_{it}$$

D_t are DV, which equal 1 when t = year

(b) the climate

3.

① control for both time-special and individual-special effect on yield

② reduce omitted variable bias

③ improve estimation precision.

4. start with entity - demean

$$Y_{it} - \bar{Y}_i = \beta_1 (X_{it} - \bar{X}_i) + \gamma_t - \bar{\gamma} + (u_{it} - \bar{u}_i)$$

$$\Rightarrow (Y_{it} - \bar{Y}_i) - (\overline{Y_{i\cdot} - \bar{Y}_i}) =$$

$$\beta_1 \left(X_{it} - \bar{X}_i - (\overline{X_{i\cdot} - \bar{X}_i})\right) + \gamma_t - \bar{\gamma} + u_{it} - \bar{u}_i - (\overline{\gamma_t - \bar{\gamma}}) - (\overline{u_{i\cdot} - \bar{u}_i})$$

start with time - demean

$$Y_{it} - \bar{Y}_t = \beta_1 (X_{it} - \bar{X}_t) + \alpha_i - \bar{\alpha} + (u_{it} - \bar{u}_t)$$

$$\Rightarrow (Y_{it} - \bar{Y}_t) - (\overline{Y_{\cdot t} - \bar{Y}_t}) =$$

$$\beta_1 (X_{it} - \bar{X}_i - (\overline{X_{\cdot t} - \bar{X}_t}) + \alpha_i - \bar{\alpha} + (u_{it} - \bar{u}_t) - (\overline{\alpha_i - \bar{\alpha}}) - (\overline{u_{\cdot t} - \bar{u}_t})$$

start with entity

$$Y_{it}^{(E,T)} = Y_{it} - \bar{Y}_i - \bar{Y}_t + \bar{\bar{Y}}$$

start with time

$$Y_{it}^{(T,E)} = Y_{it} - \bar{Y}_t - \bar{Y}_i + \bar{\bar{Y}} = Y_{it}^{(E,T)}$$

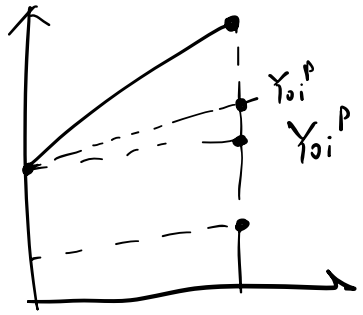
Intuitively, for balance panel datas, every entity can be observed in any time period, which ensure whether demeaning which first, we can remove the same information.

Q2

1. $DiD = \left(\frac{\sum_{i=1}^{n} D_i Y_i^1}{\sum_{i=1}^{n} D_i} - \frac{\sum_{i=1}^{n} D_i Y_i^0}{\sum_{i=1}^{n} D_i} \right) - \left(\frac{\sum_{i=1}^{n} (1-D_i) Y_i^1}{\sum_{i=1}^{n} (1-D_i)} - \frac{\sum_{i=1}^{n} (1-D_i) Y_i^0}{\sum_{i=1}^{n} (1-D_i)} \right)$

2. parallel trend assumption

3. Overestimate



because the household in Gansu works harder after the policy and irrelevant with the policy.

so the potential line of Gansu is not parallel with other province. Y_{0i}^P is under estimated

so $Did = Y_{1i}^P - Y_{0i}^P$ is over estimated

Q3

1. ① PA did not run the policy

② PA and NJ had different employment situation before the policy

2. $Empl_{it} = \beta_0 + \beta_1 treatment_{it} + \beta_2 group_i + \beta_3 treatment_{it} \times group_i + X_{it}\gamma + u_{it}$

X_{it} is a control vector

3. Parallel trend assumption

4.

```
. reg empl treatment group treatmentgroup fries
```

Source	SS	df	MS	Number of obs	=	761
				F(4, 756)	=	3.67
Model	954.870661	4	238.717665	Prob > F	=	0.0057
Residual	49138.3008	756	64.9977524	R-squared	=	0.0191
				Adj R-squared	=	0.0139
Total	50093.1715	760	65.9120677	Root MSE	=	8.0621

empl	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
treatment	-2.823965	1.33166	-2.12	0.034	-5.438155	-.2097754
group	-3.599658	1.071276	-3.36	0.001	-5.702688	-1.496629
treatmentgroup	3.566757	1.481293	2.41	0.016	.6588212	6.474694
fries	7.365167	2.908898	2.53	0.012	1.65469	13.07564
_cons	4.239513	2.617994	1.62	0.106	-.8998884	9.378914

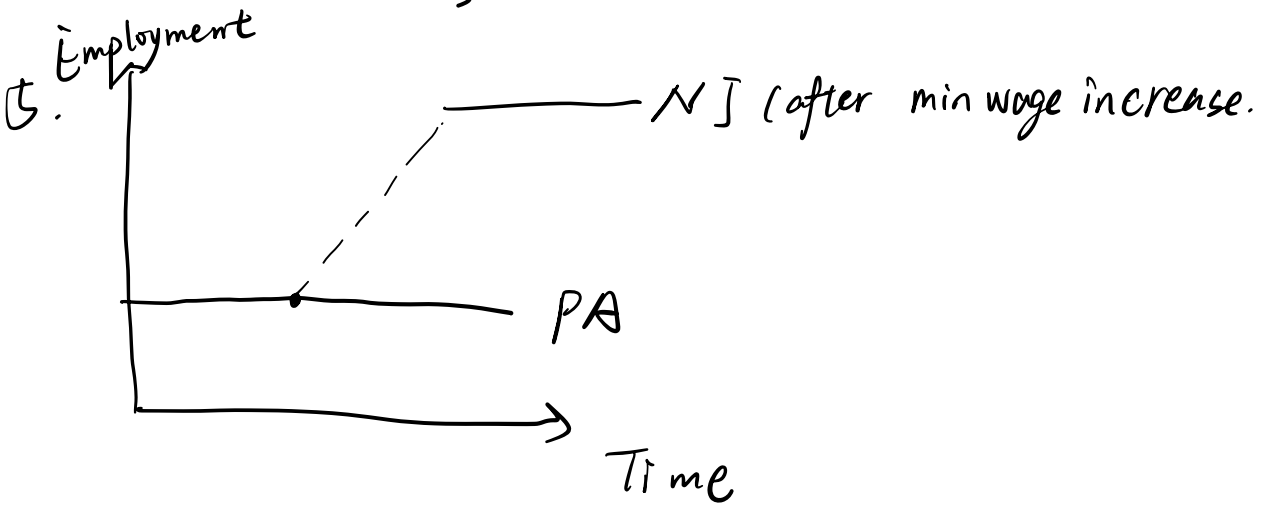
using fries as control $\beta_1 = -2.823965$

$\beta_2 = -3.599658$

$\beta_3 = 3.566757$ $\therefore$ Did $= 3.566757$

minimum wage increase from \$4.25 to 5.05\$ increase 3.566757 units of

full time employees in NJ fast food restaurants compared with fast food restaurants in PA

5. 

the divergence after the treatment occurs only because of the policy change and this treatment effect.

So there is no bias.

```
clear

cap log close

log using ps4.log,  replace

cd"/Users/hanqihang/Stata/PS4"

import delimited "/Users/hanqihang/Stata/PS4/DID_Example.csv"

replace empl = "" if empl == "NA"
replace soda = "" if soda == "NA"
replace fries = "" if fries == "NA"

destring empl, replace
destring soda, replace
destring fries, replace

gen treatmentgroup = treatment * group

reg empl treatment group treatmentgroup fries

log close
```

```
.
. cd"/Users/hanqihang/Stata/PS4"
/Users/hanqihang/Stata/PS4

.
. import delimited "/Users/hanqihang/Stata/PS4/DID_Example.csv"
(encoding automatically selected: ISO-8859-2)
(15 vars, 820 obs)

.
. replace empl = "" if empl == "NA"
(18 real changes made)

. replace soda = "" if soda == "NA"
(30 real changes made)

. replace fries = "" if fries == "NA"
(45 real changes made)

.
. destring empl, replace
empl: all characters numeric; replaced as double
(18 missing values generated)

. destring soda, replace
soda: all characters numeric; replaced as double
(30 missing values generated)

. destring fries, replace
fries: all characters numeric; replaced as double
(45 missing values generated)

.
. gen treatmentgroup = treatment * group

.
. reg empl treatment group treatmentgroup fries

      Source |       SS           df       MS      Number of obs   =       761
-------------+----------------------------------   F(4, 756)       =      3.67
       Model |  954.870661         4  238.717665   Prob > F        =    0.0057
    Residual |  49138.3008       756  64.9977524   R-squared       =    0.0191
-------------+----------------------------------   Adj R-squared   =    0.0139
       Total |  50093.1715       760  65.9120677   Root MSE        =    8.0621

------------------------------------------------------------------------------
        empl | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
   treatment |  -2.823965    1.33166    -2.12   0.034    -5.438155   -.2097754
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treatmentgroup |   3.566757   1.481293     2.41   0.016     .6588212    6.474694
       fries |   7.365167   2.908898     2.53   0.012      1.65469    13.07564
       _cons |   4.239513   2.617994     1.62   0.106    -.8998884    9.378914
------------------------------------------------------------------------------

.
. log close
```